

Fecal Microbiota Transplantation in Parkinson Disease

A Systematic Review, Meta-Analysis, and Meta-Regression

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Abstract

Background and Objectives

Parkinson disease (PD) is a progressive neurodegenerative disorder increasingly linked to gut microbiota dysbiosis, which may influence disease mechanisms and symptom expression. Fecal microbiota transplantation (FMT) targets the gut-brain axis, but clinical evidence remains inconsistent. This study aimed to evaluate the efficacy and safety of FMT in PD.

Methods

We conducted a systematic review and meta-analysis following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, with protocol registration in International Prospective Register of Systematic Reviews (CRD420251142846). MEDLINE, Embase, and the Cochrane Library were searched from inception through September 2025. Randomized controlled trials (RCTs) and observational studies enrolling adults with mild-to-moderate PD who received FMT through any administration route were eligible. The primary outcome was motor function assessed by the Unified Parkinson's Disease Rating Scale (UPDRS) part III. Secondary outcomes included UPDRS part II, quality of life (Parkinson's Disease Questionnaire-39 [PDQ-39]), constipation severity (Wexner score), and adverse events. Random-effects models pooled effect estimates with 95% CIs, and exploratory meta-regression assessed follow-up duration, publication year, and sample size.

Results

Eight studies (5 RCTs and 3 observational studies) including 220 participants were analyzed. The mean age ranged from approximately 60 to 70 years, and women comprised about 40% of participants. FMT was associated with significant improvement in motor function (UPDRS part III: mean difference [MD] −9.67, 95% CI −16.81 to −2.53) and constipation severity (Wexner score: MD −3.91, 95% CI −7.68 to −0.13). Improvements in UPDRS part II and PDQ-39 were observed at 12 weeks but not sustained at 24 weeks. In RCT-only analyses, UPDRS part III improvement remained significant (MD −6.82, 95% CI −11.23 to −2.40), whereas other outcomes were not consistently significant. Meta-regression indicated that longer follow-up was associated with greater improvement in UPDRS part II ($p = 0.043$). FMT was generally well tolerated; however, gastrointestinal adverse events were more frequent in the FMT group (risk ratio 3.12, 95% CI 1.14–8.53), predominantly mild to moderate.

Discussion

FMT may provide short-term improvements in motor and gastrointestinal symptoms in PD, but effects appear transient. Small sample sizes, heterogeneity, and limited follow-up restrict conclusions, underscoring the need for larger randomized trials. Pooled estimates reflected evidence from observational studies and should be interpreted cautiously.

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Supplementary Material

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Glossary

AE = adverse event; **FMT** = fecal microbiota transplantation; **GI** = gastrointestinal; **LEDD** = levodopa equivalent daily dose; **MD** = mean difference; **PD** = Parkinson's disease; **PDQ-39** = Parkinson's Disease Questionnaire-39; **RCT** = randomized controlled trial; **RoB 2** = Cochrane Risk of Bias 2.0; **ROBINS-I** = Risk of Bias in Nonrandomized Studies of Interventions; **RR** = risk ratio; **UPDRS** = Unified Parkinson's Disease Rating Scale.

Introduction

Parkinson disease (PD) is the second most common progressive neurodegenerative disorder, affecting an estimated 1%–3% of individuals older than 65 years, and prevalence has increased substantially worldwide, rising by about 274% since 1990.^{1–3} This condition is primarily linked to reduced dopamine levels in the brain, which lead to the classical motor manifestations of bradykinesia, rigidity, tremor, and postural instability.⁴ Beyond these classical features, however, patients frequently present a wide range of nonmotor manifestations that often precede motor symptoms and have a substantial impact on quality of life. Among them, gastrointestinal (GI) dysfunction is particularly prominent, affecting nearly 50% of patients after PD diagnoses.^{5–8}

Over the past decade, accumulating evidence has reinforced the association between GI dysfunction and alterations in the gut microbiota, suggesting that these changes may either contribute to the pathogenesis of PD or represent early prodromal manifestations.^{4,9,10} According to Braak's hypothesis, misfolded α -synuclein may initially arise in the enteric nervous system and subsequently propagate to the CNS through the vagus nerve, providing a mechanistic explanation for the early onset of GI symptoms.^{4,11} This hypothesis points to gut microbiota as a key contributor to disease onset and progression, opening promising avenues for therapeutic strategies that modulate the gut-brain axis suggesting that the condition is no longer viewed exclusively as a “brain-first” disorder.

In this context, fecal microbiota transplantation (FMT) is a therapeutic intervention designed to restore a healthy intestinal microbiome by transferring stool from a healthy donor to the patient's GI tract.¹² This approach has been safely and effectively applied in recurrent *Clostridioides difficile* infections and has also been investigated in other disorders associated with gut dysbiosis, including inflammatory bowel disease and chronic pouchitis.^{13–15} Given its potential to modulate the gut microbiota, FMT has emerged as a promising approach to improve PD-related symptoms.^{12,16}

However, the existing evidence remains limited and heterogeneous and no firm conclusions can yet be drawn regarding its efficacy and safety. Therefore, this systematic review and meta-analysis aims to evaluate the therapeutic role of FMT in patients with PD.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

This study is a systematic review and meta-analysis based exclusively on data extracted from previously published studies. Therefore, institutional review board approval and written informed consent were not required. The review protocol was registered in the International Prospective Register of Systematic Reviews (CRD420251142846). The study was conducted in accordance with the methodological framework of the Cochrane Handbook for Systematic Reviews of Interventions and adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.^{17,18}

Search Strategy

A comprehensive literature search was performed in MEDLINE (through PubMed), Embase, and the Cochrane Library from database inception to September 2025. The search strategy combined controlled vocabulary and free-text terms related to “Fecal Microbiota Transplantation” and “Parkinson disease” using Boolean operators, as follows: (“Fecal Microbiota Transplant” OR “Intestinal Microbiota Transfer” OR “Fecal Bacteriotherapy” OR “FMT” OR “Stool Transplantation”) AND (“Parkinson's disease” OR “Parkinson disease”). Two reviewers independently screened all titles, abstracts, and full-text articles to determine eligibility.

Eligibility Criteria

Eligible studies were original human investigations enrolling participants diagnosed with PD who received FMT, regardless of administration route or donor source. Studies were required to include a comparator (placebo, baseline, or standard care) and to report at least 1 clinically relevant outcome, such as motor function (Unified Parkinson's Disease Rating Scale [UPDRS] part II or part III), nonmotor symptoms, quality of life (Parkinson's Disease Questionnaire-39 [PDQ-39]), constipation severity (Wexner score), or adverse events (AEs). Both randomized controlled trials (RCTs) and observational studies were eligible, provided that participants had mild to moderate disease with a duration of 10 years or less and that sufficient data were available to estimate effect sizes.

Studies were excluded if they were nonoriginal (such as reviews, theses, letters, or conference abstracts), conducted in animals, published in languages other than English, or involved additional interventions capable of confounding results, including probiotics, intensive physiotherapy, or

surgical procedures. To avoid duplication, the research team cross-checked all included records for overlapping populations, registries, and recruitment periods. In cases of overlap, the most comprehensive and recent data set with complete outcome reporting was retained according to a prespecified hierarchy.

Data Extraction and Outcomes

Two reviewers independently extracted data using a standardized spreadsheet, recording information on study identifiers, country of origin, design (single or multicenter), sample size, demographic characteristics, disease duration, Hoehn-Yahr stage, levodopa equivalent daily dose (LEDD), follow-up duration, and AEs. Disagreements were resolved by discussion and consensus, with input from a more experienced researcher when necessary.

Efficacy outcomes were defined as changes in motor and symptomatic measures associated with PD. The primary end point was motor function, assessed through the UPDRS part III. Secondary end points included activities of daily living (UPDRS part II), quality of life (PDQ-39), and constipation severity (Wexner score). Safety outcomes comprised the incidence of AEs, particularly GI manifestations such as nausea, constipation, abdominal pain, diarrhea, bloating, and flatulence.

Subgroup Analysis and Meta-Regression

Subgroup analyses were conducted to explore whether treatment effects varied according to follow-up duration, categorized as short-term (≤ 4 weeks), intermediate (8–12 weeks), or long-term (≥ 24 weeks). Meta-regression analyses were performed to evaluate the influence of follow-up time, year of publication, and total sample size on efficacy outcomes, assessing whether these clinical and methodological factors explained the observed heterogeneity among studies.

Quality Assessment

The risk of bias was assessed independently by 2 reviewers using the Cochrane Risk of Bias 2.0 (RoB 2) tool for RCTs and the Risk of Bias in Nonrandomized Studies of Interventions (ROBINS-I) tool for observational studies.^{19,20} The RoB 2 tool evaluates bias in 5 domains, while ROBINS-I addresses 7, including confounding, participant selection, classification of interventions, and missing data. Studies were classified as having low, moderate, or serious risk of bias. Disagreements between reviewers were resolved through consensus, with assistance from a senior researcher when required.

The certainty of evidence was appraised using the Grading of Recommendations Assessment, Development, and Evaluation framework, classifying the overall confidence as high, moderate, low, or very low based on risk of bias, inconsistency, indirectness, and imprecision.

Statistical Analyses

All statistical analyses were performed using R version 4.3.0 (R Core Team, 2023) with the meta package. Pooled mean

differences (MDs) for continuous outcomes and pooled proportions for single-arm data were calculated with 95% CIs using a random-effects model based on the restricted maximum likelihood estimator. Between-study heterogeneity was assessed using the Cochran Q test and the I^2 statistic, with values above 40% indicating substantial heterogeneity. Sensitivity analyses were performed using a leave-one-out approach to test result robustness. Meta-regression analyses were planned to explore potential study-level moderators; however, given the limited number of available studies, these analyses were prespecified as exploratory. A p value < 0.05 was considered statistically significant. Owing to the limited number of studies (< 10 per analysis), the Egger regression test for publication bias was not conducted.

Data Availability

All data used in this analysis are published data. Data extracted from included studies, data used for all analyses, and analytic code are all available on request from the lead author.

Results

Search Results and Study Characteristics

A total of 723 records were identified through database searching: 135 from PubMed, 458 from Embase, and 130 from Web of Science. After the removal of 239 duplicates, 484 unique records were screened by title and abstract. Eighteen articles were selected for full-text review, of which 8 met the inclusion criteria and were ultimately included in the meta-analysis.^{21–28} The selection process is summarized in Figure 1.

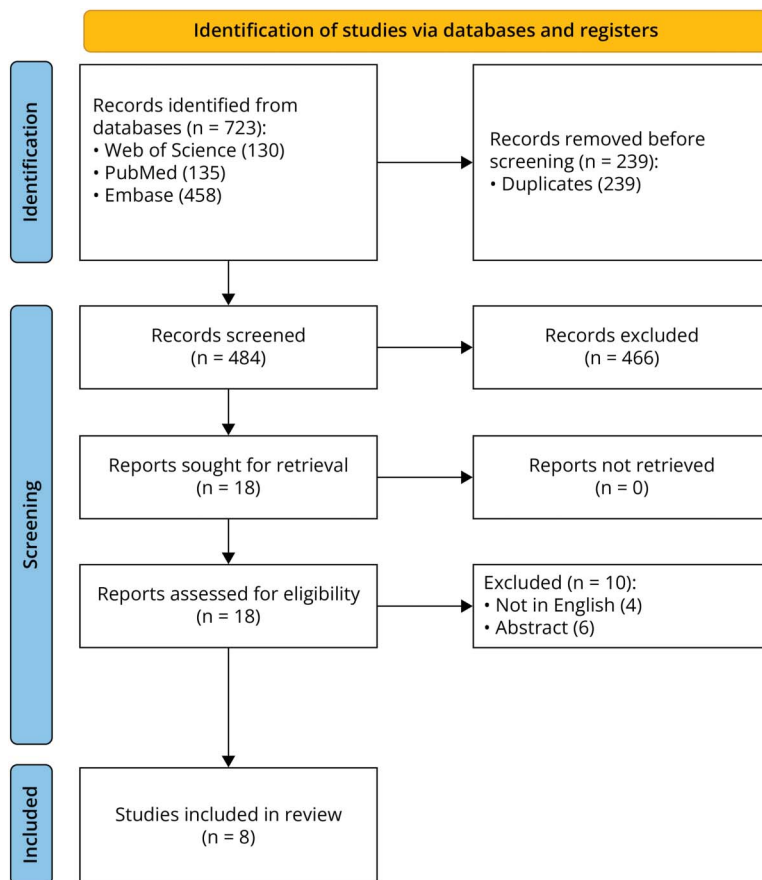
Baseline Characteristics of the Included Studies and Patients

Altogether, the 8 studies encompassed 220 patients, including 5 RCTs and 3 observational studies. The RCTs included 102 participants in the FMT group and 85 in the placebo group, while the observational studies enrolled 33 patients treated with FMT, yielding a total of 135 participants receiving FMT and 85 receiving placebo. China was the most represented country, contributing half of the included studies.^{22,25,27,28} The mean age of participants ranged from 60.5 to 69.5 years, with males comprising 57.8% of the FMT group and 63.5% of the placebo group. Only 1 nonrandomized study did not report disease duration²³; in the remaining studies, the median duration ranged from 4.2 to 7.2 years. Four studies reported the LEDD,^{21,23,24,26} and constipation was described in 4 publications,^{21,22,25,27} affecting 43.5% of patients in the FMT group and 70.5% in the placebo group. Detailed baseline characteristics of all included studies are summarized in Table 1.

Efficacy Outcomes

FMT was associated with a significant improvement in constipation severity, as demonstrated by a reduction in Wexner scores at 12 weeks compared with baseline (MD -4.04 ; 95% CI -6.72 to -1.35). Although the 24-week effect was not

Figure 1 PRISMA Flow Diagram Summarizing the Study Selection Process



PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

statistically significant (MD -3.85 ; 95% CI -11.53 to 3.83), the overall pooled analysis across all time points continued to favor post-FMT values (MD -3.91 ; 95% CI -7.68 to -0.13 ; $I^2 = 95.6\%$) (Figure 2). When the analysis was restricted to RCTs, the effect was not significant (MD 0.96 ; 95% CI -1.65 to 3.57 ; $I^2 = 73.6\%$) (eFigure 1).

Regarding the UPDRS part II (activities of daily living), FMT produced a significant improvement at 12 weeks (MD -4.22 ; 95% CI -7.02 to -1.42 ; $I^2 = 51.9\%$), although the benefit was not sustained at 24 weeks (MD -4.27 ; 95% CI -15.28 to 6.75 ; $I^2 = 98.8\%$). Between-group differences across follow-up durations were not statistically significant ($p = 0.9868$). The pooled effect across all time points did not reach significance (MD -4.20 ; 95% CI -8.86 to 0.46 ; $I^2 = 95.1\%$) (Figure 3). In RCTs only, FMT demonstrated a consistent reduction in UPDRS part II scores vs placebo at 12 weeks (MD -4.13 ; 95% CI -6.43 to -1.83 ; $I^2 = 0\%$) (eFigure 2).

Analysis of the PDQ-39 (quality of life) showed significant overall improvement following FMT (MD -13.95 ; 95% CI -24.82 to -3.08 ; $I^2 = 89.3\%$), although this effect was not sustained at 12 weeks (MD -4.85 ; 95% CI -12.76 to 3.06 ; $I^2 = 35.6\%$) (Figure 4). In RCTs only, FMT showed no significant

difference between post-FMT and pre-FMT groups at 12 weeks (MD -4.43 ; 95% CI -14.97 to 6.11 ; $I^2 = 0\%$), nor in the pooled overall effect (MD 1.48 ; 95% CI -1.67 to 4.62 ; $I^2 = 8.6\%$) (eFigure 3).

Motor outcomes, measured by UPDRS part III, revealed significant overall improvement following FMT (MD -9.67 ; 95% CI -16.81 to -2.53 ; $I^2 = 94.4\%$), although this effect was not sustained at 24 weeks (MD -10.70 ; 95% CI -25.33 to 3.92) (Figure 5). In the RCT-only subgroup, the effect remained statistically significant (MD -6.82 ; 95% CI -11.23 to -2.40 ; $I^2 = 0\%$) (eFigure 4).

Safety Outcomes

FMT was generally well tolerated (Table 2). Across studies, GI AEs were the most frequently reported, typically mild and transient. The pooled prevalence of nausea was 10.4% (95% CI 1.1%–24.8%; $I^2 = 47.1\%$), bloating or flatulence occurred in 27.7% (95% CI 4.0%–59.7%; $I^2 = 87.6\%$), abdominal pain in 15.6% (95% CI 3.2%–32.9%; $I^2 = 48.6\%$), diarrhea in 11.2% (95% CI 3.7%–21.0%; $I^2 = 4.1\%$), and constipation in 9.8% (95% CI 0%–34.3%; $I^2 = 81.8\%$). The pooled incidence of any GI AE was 62.6% (95% CI 46.7%–77.4%; $I^2 = 38.9\%$) (eFigures 5–10). Compared with placebo, FMT was not

Table 1 Baseline Characteristics of Included Studies and Participants

Study (year)	Country	Design	Sex (M/F)	Age, y Mean $\pm$ SD or median (range/IQR)	Disease duration, y Mean $\pm$ SD or median (range/IQR)	Hoehn-Yahr stage Mean $\pm$ SD or median (IQR)	UPDRS III Mean $\pm$ SD or median (IQR)	LEDD, mg Mean $\pm$ SD or median (range/IQR)	Constipation (n)
Bruggeman et al. (2024) ²¹	Belgium	RCT	FMT: 15/7 Placebo: 14/10	FMT: 61 $\pm$ 1.1 Placebo: 60.5 $\pm$ 0.7	FMT: 4.2 $\pm$ 0.7 Placebo: 4.4 $\pm$ 0.7	NR	FMT: 40.3 (2.7) ^a Placebo: 37.1 (2.5) ^a	FMT: 383 $\pm$ 53 Placebo: 431 $\pm$ 51	FMT: 14 Placebo: 15
Cheng et al. (2023) ²²	China	RCT	FMT: 15/12 Placebo: 17/10	FMT: 60.52 $\pm$ 8.68 Placebo: 62.63 $\pm$ 8.41	NR	NR	FMT: 24.74 (10.58) Placebo: 29.59 (13.07)	NR	FMT: 23 Placebo: 24
DuPont et al. (2023) ²⁴	USA	RCT	FMT: 5/3 Placebo: 4/0	FMT: 68.5 (14) Placebo: 68 (56–74)	FMT: 2 (1–3) Placebo: 3 (2–10)	FMT: 1 (1–2) Placebo: 2.2 (1–2.5)	FMT: 16 (13–21) Placebo: 26 (18–42)	FMT: 446 (114–900) Placebo: 645 (400–792)	NR
Scheperjans et al. (2024) ²⁶	Finland	RCT	FMT: 16/14 Placebo: 9/6	FMT: 66 (59.25–69.75) Placebo: 65 (52.5–70)	FMT: 5.91 (3.86–7.57) Placebo: 6.96 (2.5–8.78)	FMT: 2 (2–2) Placebo: 2 (2–2)	OFF: 29.5 (21.75–37.75) ON: 18.5 (13.75–23.75) Placebo OFF: 20 (18.5–31.5) ON: 14 (11.5–21.5)	FMT: 670 (405–1,031.12) Placebo: 712 (400–946.74)	NR
Wang et al. (2025) ²⁷	China	RCT	FMT: 8/7 Placebo: 10/5	FMT: 67 (62.70) ^b Placebo: 68 (59.72) ^b	FMT: 64 (56.65) ^b Placebo: 63 (51.70) ^b	NR	NR	NR	FMT: 13 Placebo: 12
Kuai et al. (2021) ²⁵	China	Observational	7/4	62.45 $\pm$ 13.08	7.18 $\pm$ 3.25	2.27 (1–3)	NA	NR	11
De Sciscio et al. (2025) ²³	Australia	Observational	8/4	69.5 $\pm$ 4.3	NR	NR	OFF: 28.3 (12.5) ON: 12.7 (8.3)	684 $\pm$ 431	NR
Xue et al. (2020) ²⁸	China	Observational	7/3	60.9 $\pm$ 7.74	5.4 $\pm$ 3.5	3	43.60 $\pm$ 14.64	475 $\pm$ 207.6	NR

Abbreviations: FMT = fecal microbiota transplantation; IQR = interquartile range; LEDD = levodopa equivalent daily dose; NR = not reported; RCT = randomized controlled trial; SEM = standard error of the mean; UPDRS = Unified Parkinson's Disease Rating Scale.

^a Reported as SEM.

^b Reported as median (IQR).

associated with a statistically significant increase in AEs (risk ratio [RR] 1.28; 95% CI 0.72–2.29; $I^2 = 52.8\%$), diarrhea (RR 3.40; 95% CI 0.617–18.83; $I^2 = 0\%$), nausea (RR 0.75; 95% CI 0.10–5.49; $I^2 = 0\%$), bloating or flatulence (RR 1.06; 95% CI 0.23–4.87; $I^2 = 0.9\%$), and in constipation (RR 0.80; 95% CI 0.37–1.73; $I^2 = 0\%$) (eFigures 11–15).

Meta-Regression Analysis

Owing to the small number of included studies, the meta-regression analyses were underpowered due to limited statistical power and should be interpreted as exploratory in nature. For UPDRS part III, PDQ-39, and Wexner scores, follow-up duration, publication year, and total sample size were assessed. Longer follow-up was significantly associated with greater improvement in UPDRS part II scores ($\beta = 0.214$; $p = 0.043$), while more recent publications ($\beta = -2.69$; $p = 0.012$) and larger sample sizes ($\beta = 0.168$; $p = 0.012$) were significantly related to greater reductions in Wexner scores. No other significant associations were observed. Residual heterogeneity was negligible for PDQ-39 ($I^2 = 0\%$) and moderate to high for UPDRS part II and Wexner ($I^2 = 66\%$ – 80%).

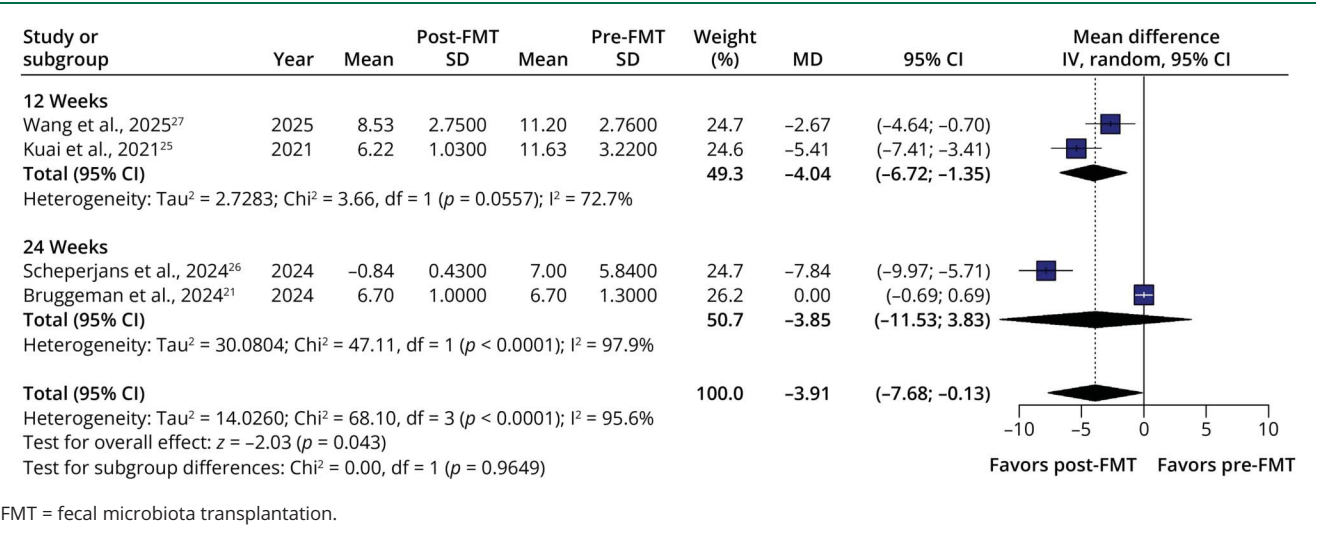
Sensitivity Analysis

Leave-one-out sensitivity analyses confirmed the robustness of the pooled results. For the Wexner score, pooled effects ranged from -5.28 to -2.60 , with Bruggeman et al. exerting the greatest influence on overall heterogeneity ($I^2 = 83.7\%$ – 97.0%). For UPDRS part III, results ranged from -6.68 to -11.41 ($I^2 = 41.1\%$ – 89.6%), with Scheperjans et al. contributing most to variability. Similarly, for UPDRS part II, estimates ranged from -5.56 to -2.81 ($I^2 = 80.0\%$ – 96.0%), while for PDQ-39, values ranged from -17.97 to -10.77 ($I^2 = 40.9\%$ – 91.5%). In RCT-only subgroups, heterogeneity for UPDRS part II and PDQ-39 was eliminated ($I^2 = 0\%$) when either Bruggeman et al. or Wang et al. was excluded. For AEs in RCTs subgroup, RR ranged from 0.91 to 1.90, omitting DuPont et al. Bruggeman et al. resolved the heterogeneity ($I^2 = 0\%$). All funnel plots, leave-one-out sensitivity analyses, and Baujat plots are presented in eFigures 16–28.

Quality Assessment and Certainty of Evidence

Quality assessment revealed variable methodological rigor across studies. Among the 3 nonrandomized studies assessed with ROBINS-I, 2 were judged to present a critical risk of bias

Figure 2 Forest Plot for the Effect of FMT on Wexner Constipation Scores at 12 Weeks, 24 Weeks, and Pooled Across All Time Points



and 1 a serious risk. Among the 5 RCTs evaluated with RoB 2, 4 were classified as having a low risk of bias and 1 as showing some concerns. These findings indicate an overall moderate level of certainty in the available evidence, with methodological variability contributing to residual heterogeneity (eFigure 29).

Among RCT-only analyses (Wexner score, and UPDRS part III), the certainty of evidence was downgraded by 1 level due to imprecision, as only 2 studies contributed to some outcomes, resulting in wide CIs and precluding meaningful sensitivity analyses. For analyses combining randomized and

nonrandomized studies (Wexner score, UPDRS part II, UPDRS part III, and PDQ-39), the certainty was also rated as moderate, downgraded by 1 level due to risk of bias, as several included nonrandomized studies were judged to have moderate methodological limitations according to ROBINS-I (eTable 1).

Discussion

This systematic review and meta-analysis integrated findings from 8 eligible studies: 5 RCTs and 3 observational cohorts,

Figure 3 Forest Plot for the Effect of FMT on UPDRS Part II (Activities of Daily Living) at 12 Weeks, 24 Weeks, and Pooled Across All Time Points

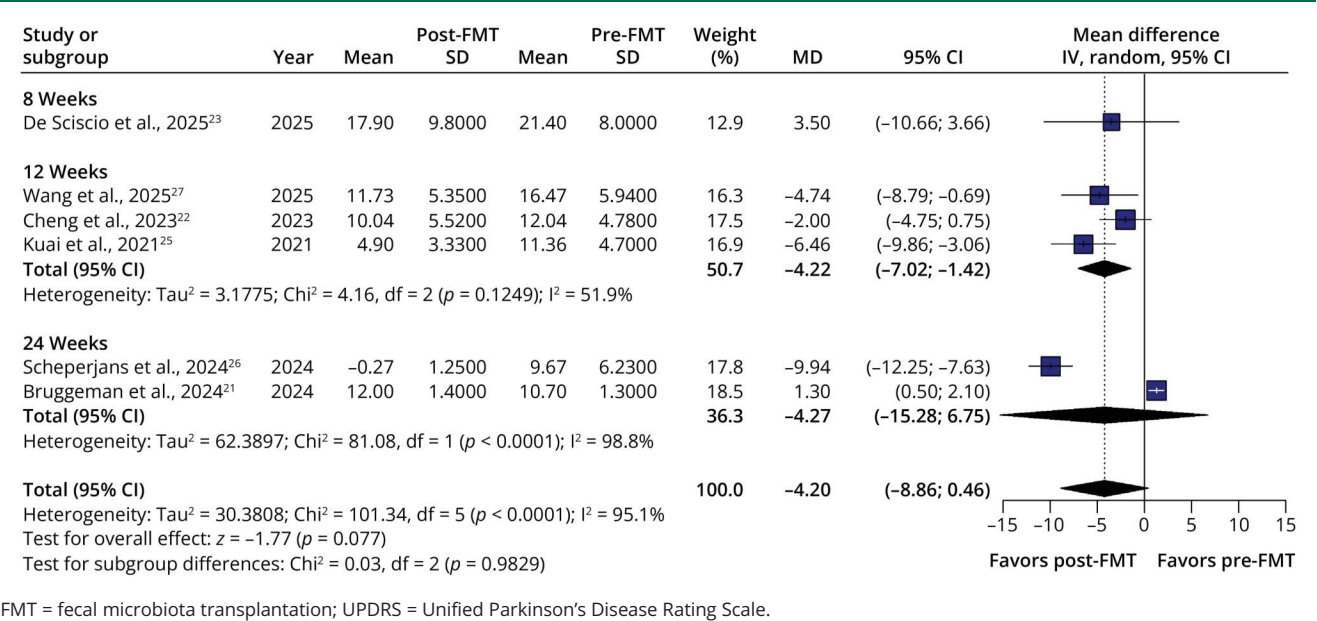
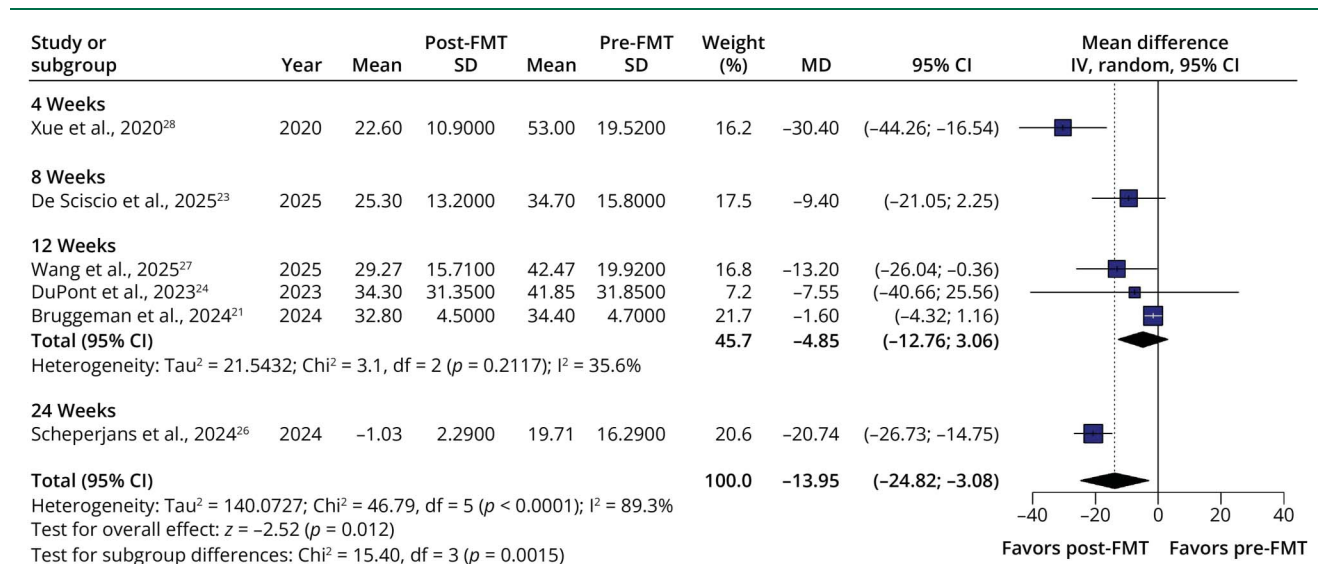


Figure 4 Forest Plot for the Effect of FMT on PDQ-39 (Quality of Life) at 12 Weeks and Pooled Overall Effect



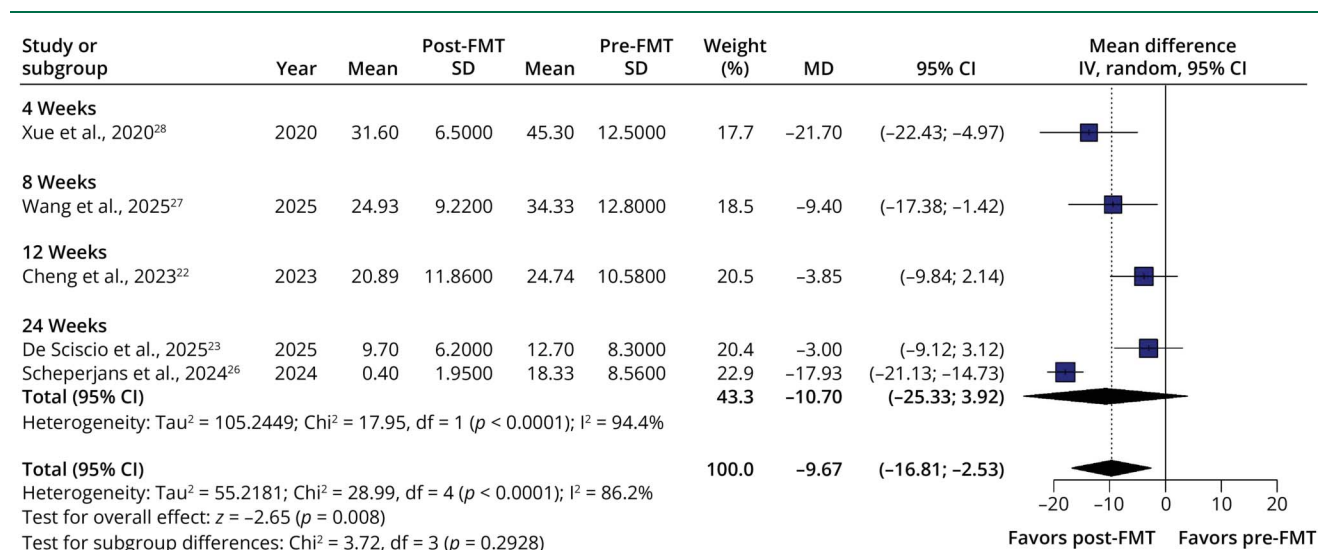
FMT = fecal microbiota transplantation; PDQ-39 = Parkinson's Disease Questionnaire-39.

comprising a total of 220 patients with PD who underwent FMT. Pooled estimates, largely driven by nonrandomized studies, suggest that FMT may provide short-term improvements in constipation and motor and nonmotor symptoms. However, these effects were not sustained at longer follow-up, and heterogeneity was substantial across most analyses. When restricted to RCTs, several associations lost statistical significance, with consistent benefits remaining primarily for UPDRS part II and UPDRS part III at 12 weeks and with low heterogeneity. Regarding quality of life, pooled analyses indicated an overall improvement in PDQ-39 scores; however,

this effect was not sustained at short-term follow-up and was not confirmed in RCT-only analyses. Regarding safety, FMT was overall well tolerated, with mostly mild and transient GI AEs, and was not associated with a statistically significant increase in AEs compared with placebo.

When considering all included studies together, FMT produced significant improvements in UPDRS part III, UPDRS part II, and Wexner score reinforcing the hypothesis that modulation of the gut microbiota can influence both motor and nonmotor manifestations of PD through the gut-brain

Figure 5 Forest Plot for the Effect of FMT on UPDRS Part III (Motor Function) at 12 Weeks, 24 Weeks, and Pooled Across All Time Points



FMT = fecal microbiota transplantation; UPDRS = Unified Parkinson's Disease Rating Scale.

Table 2 Summary of Pooled Analyses of AEs Following FMT

AE	Pooled analysis (RCTs + cohorts)			RCTs only		
	No. of studies	Proportion (95% CI)	I ² (%)	No. of studies	RR (95% CI)	I ² (%) / p value
Nausea	5	10.39 (1.10–24.81)	47.1	2	0.75 (0.10–5.49)	0/0.777
Bloating/flatulence	6	27.72 (4.03–59.69)	87.6	3	1.06 (0.23–4.87)	0.9/0.937
Abdominal pain	5	15.60 (3.23–32.90)	48.6	2	1.69 (0.24–11.71)	0/0.596
Diarrhea	6	11.17 (3.71–21.03)	4.1	3	3.40 (0.61–18.83)	0/0.161
Constipation	5	9.79 (0.00–34.28)	81.8	4	0.80 (0.37–1.73)	0/0.565
Any gastrointestinal AE	3	62.62 (46.69–77.38)	38.9	4	1.28 (0.72–2.29)	52.8/0.400

Abbreviations: AE = adverse event; FMT = fecal microbiota transplantation; RCT = randomized controlled trial; RR = risk ratio.

axis.²⁹ However, the progressive decline of these effects by 24 weeks suggests that the benefits may be temporary. While the underlying reasons for this attenuation remain unclear, possible and speculative explanations include recolonization by dysbiotic microbial communities or the absence of sustained maintenance protocols. These hypotheses should be interpreted as a possible explanation rather than a definitive conclusion. In the subgroup of RCTs, the effect persisted only for motor outcomes, indicating that FMT may serve as an adjunctive symptomatic therapy rather than a disease-modifying intervention.³⁰

These findings are consistent with the evolving view of PD as a multifactorial condition that may arise through distinct mechanistic pathways, most notably the proposed “body-first” and “brain-first” subtypes. Within this conceptual framework, patients with early enteric α -synuclein deposition and prominent GI involvement may represent a subgroup more likely to respond to microbiota-targeted interventions, whereas those with predominantly CNS pathology may derive comparatively limited therapeutic benefit.^{31,32} This pathophysiologic heterogeneity, combined with variation in disease stage, symptom profile, and baseline microbiota composition, likely contributes to the inconsistent results observed across studies. Future trials should therefore incorporate phenotype-based stratification and biomarker-guided screening to determine which subgroups of patients are most likely to benefit from FMT.

From a broader therapeutic standpoint, these observations should be interpreted in the context of current PD management, which remains primarily symptomatic. Despite meaningful advances in dopaminergic replacement strategies and surgical interventions such as deep brain stimulation, no therapy has yet demonstrated disease-modifying efficacy. Recently approved formulations, such as continuous subcutaneous infusions of foslevodopa/foscarbidopa and ND0612, represent important pharmacologic innovations designed to enhance dopaminergic stability and reduce motor fluctuations but still do not alter disease progression.^{33–35}

Within this therapeutic landscape, microbiota-directed approaches like FMT stand out as novel strategies that act on upstream mechanisms implicated in neurodegeneration, including neuroinflammation, oxidative stress, and the propagation of misfolded α -synuclein, through restoration of intestinal microbial homeostasis.³⁶

Mechanistically, the rationale for FMT in PD is supported by experimental and translational evidence. Animal models have shown that gut microbiota composition influences dopaminergic neuron survival, α -synuclein aggregation, and microglial activation.^{37,38} Transplanting fecal microbiota from healthy donors into PD model animals improves motor performance, restores neurotransmitter balance, and attenuates neuroinflammation, effects largely mediated by modulation of the Toll-like receptor 4/tumor necrosis factor α signaling pathway.^{39–41} Clinically, early human trials and case series have reported parallel findings, demonstrating improvements in GI function and motor symptoms following FMT administered through colonoscopy, nasojejunal tube, or oral capsules, with AEs that were mild and self-limiting.⁴²

Safety analyses indicate that FMT is generally well tolerated in PD populations. The most frequent AEs in studies were GI, including bloating, nausea, abdominal pain, diarrhea, and constipation, consistent with findings from FMT studies in other disorders.³⁰ Although the pooled analysis did not reveal an increased overall incidence of AEs in the FMT group compared with placebo (RR 1.28; 95% CI 0.72–2.29), diarrhea (RR 3.40; 95% CI 0.617–18.83), and constipation (RR 0.80; 95% CI 0.37–1.73), the frequency of mild-to-moderate GI symptoms underscores the need for careful monitoring, particularly in older or frailer patients and those with preexisting GI conditions. Importantly, these transient symptoms are largely offset by the observed short-term improvements in motor and nonmotor outcomes, suggesting an overall favorable balance between clinical benefit and risk.

Across included studies, heterogeneity in FMT administration may partly account for variability in clinical outcomes.

Delivery routes ranged from nasojejunal or nasoduodenal infusion to colonic administration and oral capsules, which may differentially affect microbial engraftment and symptom domains. Preparation methods also varied, with studies using fresh FMT more frequently reporting sustained motor and nonmotor benefits, whereas cryopreserved or lyophilized preparations may be associated with reduced microbial viability and less consistent effects. In addition, dosing strategies differed from single administrations to induction protocols with maintenance dosing, with evidence suggesting that benefits may wane without repeated exposure. Donor-related factors further contribute to variability, as engraftment appears donor-dependent, raising the possibility that specific high-diversity donor profiles are required for optimal therapeutic response in PD. The transient nature of these effects underscores the need for standardized maintenance protocols and controlled evaluations of repeated dosing regimens to sustain clinical efficacy.^{29,43}

Meta-regression findings provided further insight into factors influencing treatment response. Longer follow-up duration was significantly associated with greater improvements in UPDRS part II scores, suggesting a possible delayed symptomatic effect on daily living activities. For the Wexner score, there was a significant association between publication year and treatment effect, with more recent studies showing greater reductions, possibly reflecting advances in FMT methodology. Larger sample sizes were also correlated with more pronounced improvements in constipation severity, emphasizing how study design can shape observed outcomes.

This review also carries inherent limitations that should be acknowledged. A substantial portion of the pooled evidence derived from nonrandomized studies at serious or critical risk of bias, particularly due to confounding and deviations from intended interventions. These limitations may overestimate treatment effects. Notably, several outcomes lost statistical significance in RCT-only analyses, suggesting that the overall pooled benefits may be partially driven by higher-risk observational data and should therefore be interpreted cautiously. The number of eligible studies was small, and most involved limited sample sizes, which constrains statistical power and generalizability. Significant clinical and methodological heterogeneity, particularly regarding administration route, frequency, and concomitant therapies, likely influenced pooled estimates. Furthermore, the absence of validated biomarkers for GI involvement in PD precluded stratification by disease subtype, limiting interpretation of differential responsiveness. Short follow-up durations restricted assessment of long-term outcomes, and inconsistencies in AE reporting reduced the accuracy of safety estimates.

Future research should prioritize large, multicenter, randomized, double- or triple-blind placebo-controlled trials with standardized FMT protocols and extended follow-up (≥ 24 months) to adequately assess durability of response and potential disease-modifying effects.⁴⁴ To reduce methodological

heterogeneity and improve cross-trial comparability, harmonization of administration strategies—including delivery route, preparation method, and dosing regimens incorporating induction and maintenance phases—will be essential. Patient selection and stratification also warrant refinement; targeting biologically informed subgroups, such as individuals with prominent GI dysfunction or “body-first” PD phenotypes, and incorporating baseline microbiota profiling to identify dysbiosis may enhance signal detection. Finally, careful control of key confounders, including dietary intake and bowel preparation effects, is critical to ensure robust and interpretable results.

To the best of our knowledge, this is the first meta-analysis to provide a comprehensive synthesis of the available evidence on FMT in PD, integrating both randomized and non-randomized studies to evaluate effects on motor and non-motor outcomes. While FMT appears generally safe, several outcomes lose statistical significance when analyses are restricted to RCTs or longer-term follow-up, highlighting that the current evidence remains preliminary. These findings underscore the need for further well-designed and adequately powered RCTs, including studies evaluating standardized FMT protocols, maintenance or repeat dosing strategies, longer follow-up periods, and patient stratification based on clinical or biological markers.

Author Contributions

M.A. Oliveira Machado Pereira: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. G. Caruso Novaes Tudella: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A.U. Klostermann: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. V.A. Ohanesian: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. O.R. Gonçalves: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. R.O. da Silva: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.V. Andrade Fernandes: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A. Ahmed: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A.B.B. Alencar: major role in the acquisition of data. S.T.A. de Moraes: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. K.J. Almeida: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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